Probability & Statistics

UNIT V Testing of Hypothesis

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Procedure

1. Null hypothesis
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1. Null hypothesis H_0 is defined.
2. Alternative hypothesis H_1 is defined after carefully study of the problem and also the nature of the test **(one tailed or two tailed)** is decided.
3. LOS α is fixed or taken from the problem if specified and z_α is computed.
4. $$\text{The test statistic } z = \frac{t - E(t)}{SE(t)} \text{ is computed}$$
5. Comparison is made between $|z|$ and z_α.
   If $|z| < z_\alpha$ then H_0 is accepted i.e. it is concluded that difference between t and E(t) is not significant at α% LOS.
   On the other hand,
   If $|z| > z_\alpha$ then H_0 is rejected i.e. it is concluded that difference between t and E(t) is significant at α% LOS.

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z-test

- One sample mean
- Two sample means
- One sample proportion
- Two sample proportion

Critical Points for Different Levels of Significance

Level of Significance (α)	0.10	0.05	0.01	0.005	0.002
z_c for 1-Tailed Tests	−1.28 or 1.28	−1.645 or 1.645	−2.33 or 2.33	−2.58 or 2.58	−2.88 or 2.88
z_c for 2-Tailed Tests	−1.645 and 1.645	−1.96 and 1.96	−2.58 and 2.58	−2.81 and 2.81	−3.08 and 3.08

One sample z-test

- $$\boxed{z = \frac{\bar{x} - \mu}{\left(\frac{\sigma}{\sqrt{n}}\right)}}$$
  where σ is the standard deviation of the population.
  $\bar{x}$ is sample mean, μ is the population mean, n is the sample size
- If σ is not known, we use
  $$\boxed{z = \frac{\bar{x} - \mu}{\left(\frac{s}{\sqrt{n}}\right)}}$$
  where s is the standard deviation of the sample.
  $\bar{x}$ is sample mean, μ is the population mean, n is the sample size

Question

A sample of 100 students is taken from a large population. The mean height of the students in this sample is 160 cm. Can it be reasonably regarded that, in the population, the mean height is 165cm, and the SD is 10cm?

Given:$\overline{x} =\quad ,\mu =\quad ,n =\quad ,\sigma =$

1. Null hypothesis H_0
2. Alternate Hypothesis H_1
3. Level of Significance:$Z_\alpha =$ at % LOS.
4. Test Statistic

$$z_{cal} = \frac{\overline{x} - \mu}{\left(\frac{\sigma}{\sqrt{n}}\right)}$$

5. Conclusion

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Question

A random sample of 50 items gives the mean 6.2 and standard deviation 10.24. Can it be regarded as drawn from a normal population with mean 5.4 at 5% level of significance?

Question

The mean value of a random sample of 60 items was found to be 145, with a standard deviation of 40. Find the 95% confidence limits for the population mean.

Since the population SD s too is not given, we can approximate it by the sample SDs. Therefore 95% confidence limits for m are given by

$$\frac{|\mu - \overline{x}|}{\frac{s}{\sqrt{n}}} \leq 1.96$$

Two sample mean

$$\boxed{z_{cal} = \frac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{{\sigma_1}^2}{n_1} + \frac{{\sigma_2}^2}{n_2}}}}$$

The average marks scored by 32 boys is 72 with a SD of 8, while that for 36 girls is 70 with a SD of 6. Test at 1% LOS whether the boys perform better than girls.

Given:

$\overline{x}_1 = \quad , \overline{x}_2 = \quad , n_1 = \quad ,$

$n_2 = \quad = \quad , \sigma_1 = \quad , \sigma_2 =$

1. **Null hypothesis** H_0
2. **Alternate Hypothesis** H_1

3. **Level of Significance**:$Z_\alpha = \quad$ at % LOS.

4 Test Statistic

$$\boxed{z_{cal} = \frac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{{\sigma_1}^2}{n_1} + \frac{{\sigma_2}^2}{n_2}}}}$$

5 Conclusion

Test the significance of the difference between the means of the samples, drawn from two normal populations with the same SD using the following data:

	size	mean	SD
sample 1	100	61	4
sample 2	200	63	6

Given:

$n_1 = \quad ,\bar{x}_1 = \quad ,\sigma_1 = \quad ,$

$n_2 = \quad ,,\bar{x}_2 = \quad ,\sigma_2 =$

1. **Null hypothesis** H_0
2. **Alternate Hypothesis** H_1

3. **Level of Significance**:$Z_\alpha = \quad$ at $\quad$ % LOS.

4 Test Statistic

$$\boxed{z_{cal} = \frac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{{\sigma_1}^2}{n_1} + \frac{{\sigma_2}^2}{n_2}}}}$$

5 Conclusion

A simple sample of heights of 6400 English men has a mean of 170 cm and an SD of 6.4 cm, while a simple sample of heights of 1600 Americans has a mean of 172 cm and an SD of 6.3 cm. Do the data indicate that Americans are, on the average, taller than the Englishmen?

One sample proportion

$$\boxed{z_{cal} = \frac{\overline{p} - P}{\sqrt{\frac{PQ}{n}}}}$$

Experience has shown that 20% of manufactured product is of top quality. In one day's production if 400 articles, only 50 are of top quality. Show that whether the production of the day chosen was not a representative sample or the hypothesis of 20% was wrong.

Given: $\overline{p} = \quad , P = \quad , n = \quad ,$

1. **Null hypothesis** H_0
2. **Alternate Hypothesis** H_1

3. **Level of Significance**: $Z_\alpha = \quad$ at % LOS.

4 Test Statistic

$$\boxed{z_{cal} = \frac{\overline{p} - P}{\sqrt{\frac{PQ}{n}}}}$$

5 Conclusion

In a particular area 984 girls against every 1000 boys. Can we say gender equality exist?

Two sample proportion

$$\boxed{z_{cal} = \frac{\overline{p}_1 - \overline{p}_2}{\sqrt{PQ(\frac{1}{n_1} + \frac{1}{n_2})}}}$$

$$P = \frac{n_1\overline{p}_1 + n_2\overline{p}_2}{n_1 + n_2}$$

$Q = 1 - P$

Question

In a large city A, 20% of a random sample of 900 school boys had a slight physical defect. In another large city B, 18.5 percent of a random sample of 1600 school boys had the same defect. Is the difference between the proportions significant?

Given

1. **Null hypothesis** H_0
2. **Alternate Hypothesis** H_1
3. **Level of Significance**:$Z_\alpha =$ at % LOS.

4 Test Statistic

$$P = \frac{n_1\overline{p}_1 + n_2\overline{p}_2}{n_1 + n_2}$$

$$\boxed{z_{cal} = \frac{\overline{p}_1 - \overline{p}_2}{\sqrt{PQ(\frac{1}{n_1} + \frac{1}{n_2})}}}$$

5 Conclusion

Question

Before an increase in excise duty on tea, 800 people out of a sample of 1000 were consumers of tea. After the increase in duty, 800 people were consumers of tea in a sample of 1200 persons. Find whether there is significant decrease in the consumption of tea after the increase in duty.[α=1 %]

Given

1. **Null hypothesis** H_0
2. **Alternate Hypothesis** H_1
3. **Level of Significance**:$Z_\alpha =$ at % LOS.

4 Test Statistic

$$P = \frac{n_1\overline{p}_1 + n_2\overline{p}_2}{n_1 + n_2}$$

$$\boxed{z_{cal} = \frac{\overline{p}_1 - \overline{p}_2}{\sqrt{PQ(\frac{1}{n_1} + \frac{1}{n_2})}}}$$

5 Conclusion

t-test

- One sample

$$t = \frac{\overline{x} - \mu}{\left(\frac{s}{\sqrt{n-1}}\right)}$$

- Two sample mean

$$t = \frac{\overline{x_1} - \overline{x_2}}{\sqrt{\left(\frac{n_1 s_1^2 + n_2 s_2^2}{n_1+n_2-2}\right)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

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Paired t test

$$t = \frac{\overline{d}}{\left(\frac{s}{\sqrt{n-1}}\right)}$$

$d_i = x_i - y_i$, $\overline{d} = \overline{x} - \overline{y}$

$$s^2 = \frac{1}{n}\sum_{i=1}^{n}(d_i - \overline{d})^2$$

Q.1. The mean lifetime of a sample of 25 bulbs is found as 1550 hrs with a SD of 120 hrs. The company manufacturing the bulbs claims that the average life of their bulb is 1600 hrs. Is the claim acceptable at 5 % LOS? **Given**:$\bar{x} =$,$\mu =$,$n =$,$\sigma =$

1. Null hypothesis H_0
2. Alternate Hypothesis H_1
3. Level of Significance:$t_\alpha =$ at % LOS.
4. Test Statistic

$$t = \frac{\bar{x} - \mu}{\left(\frac{s}{\sqrt{n-1}}\right)}$$

5. Conclusion

Q.2. Tests made on the breaking strength of 10 pieces of a metal gave the following results: 578, 572, 570, 568, 572, 570, 570, 572, 596 and 584 kg. Test if the mean breaking strength of the wire can be assumed as 577 kg. **Given**:$\overline{x} =$,$\mu =$,$n =$,$\sigma =$

1. Null hypothesis H_0
2. Alternate Hypothesis H_1
3. Level of Significance:$t_\alpha =$ at % LOS.
4. Test Statistic

$$t = \frac{\overline{x} - \mu}{\left(\frac{s}{\sqrt{n-1}}\right)}$$

5. Conclusion

Q.3. A certain injection administered to each of 12 patients resulted in the following increases of blood pressure, 5,2,8, -1,3,0,6, -2,1,5,0,4 Can it be concluded that the injection will be in general accompanied by an increase in BP? LOS 5%.

Q.4. A machinist is expected to make engine parts with axle diameter of 1.75 cm. A random sample of 10 parts shows a mean diameter 1.85 cm with an SD of 0.1 cm. On the basis of this sample, would you say that the work of the machinist is inferior? LOS 5%.

Given: $n_1 = \quad , \overline{x}_1 = \quad , \sigma_1 = \quad ,$

$n_2 = \quad , \overline{x}_2 = \quad , \sigma_2 =$

Q.5. Samples of two types of electric bulbs were tested for length of life and the following data were obtained:

	Size	Mean	SD
Sample I	8	1234 h	36 h
Sample II	7	1036 h	40 h

Is the difference in the means sufficient to warrant that type 1 bulbs are superior to type 2 bulbs? LOS 5%.

Given: $n_1 = \quad , \bar{x}_1 = \quad , \sigma_1 = \quad ,$

$n_2 = \quad , \bar{x}_2 = \quad , \sigma_2 =$

1. Null hypothesis H_0
2. Alternate Hypothesis H_1
3. Level of Significance: $t_\alpha =$ at % LOS.

4 Test Statistic

$$t = \frac{\overline{x_1} - \overline{x_2}}{\sqrt{\left(\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}\right)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

5 Conclusion

Q.6. The mean height and the SD height of 8 randomly chosen soldiers are 166.9 and 8.29 cm respectively. The corresponding values of 6 randomly chosen sailors are 170.3 and 8.50 cm respectively. Based on this data can we conclude that soldiers are in general shorter than sailors? LOS 1%

Given: $n_1 = \quad , \overline{x}_1 = \quad , \sigma_1 = \quad ,$
$n_2 = \quad , \overline{x}_2 = \quad , \sigma_2 =$

Q.7. Two independent samples of sizes 8 and 7 contained the following values:

Sample I	19	17	15	21	16	18	16	14
Sample II	15	14	15	19	15	18	16	

Is the difference between the sample means significant? LOS 1%

Given: $n_1 = \quad , \overline{x}_1 = \quad , s_1 = \quad ,$

$n_2 = \quad , \overline{x}_2 = \quad , s_2 =$

1. Null hypothesis H_0
2. Alternate Hypothesis H_1
3. Level of Significance: $t_\alpha =$ at % LOS.

4 Test Statistic

$$t = \frac{\overline{x_1} - \overline{x_2}}{\sqrt{\left(\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}\right)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

5 Conclusion

Q.8. The following data represent the biological values of protein from cow's milk and buffalo's milk at a certain level

Cow's milk	1.82	2.02	1.88	1.61	1.81	1.54
Buffalo's milk	2.00	1.83	1.86	2.03	2.19	1.88

Examine if the average values of protein in the two samples significantly differ at 5 % LOS.

Given: $n_1 = \quad , \overline{x}_1 = \quad , \sigma_1 = \quad ,$

$n_2 = \quad , \overline{x}_2 = \quad , \sigma_2 =$

Q.9. The following data relate to the marks obtained by 11 students in 2 tests, one held at the beginning of a year and the other at the end of the year after intensive coaching.

Test 1	19	23	16	24	17	18	20	18	21	19	20
Test 2	17	24	20	24	20	22	20	20	18	22	19

Do the data indicate that the students have benefited by coaching?

Solution: The given data relate to the marks obtained in 2 tests by the same set of students. Hence the marks in the 2 tests can be regarded as correlated and so the t-test for paired values should be used

Q.10. The sales data of an item in six shops before and after a special promotional campaign are as follows:

Shops	A	B	C	D	E	F
Before campaign	53	28	31	48	50	42
After campaign	58	29	30	55	56	45

Can the campaign be judged to be a success at 5% level of significance? [t =2.78 , Yes.]

F test

A random variable F is said to follow snedecor's F-distribution or simply
F-distribution if its probability density function is given by

$$f(F) = \frac{(\nu_1/\nu_2)^{\nu_1/2}}{\beta\left(\frac{\nu_1}{2}, \frac{\nu_2}{2}\right)} \cdot \frac{F^{\nu_1/2-1}}{\left(1+\frac{\nu_1 F}{\nu_2}\right)^{(\nu_1+\nu_2)/2}}, \quad F > 0.$$

Note *The mathematical variable corresponding to the random variable F is also taken as F. ν_1 and ν_2 used in $f(F)$ are the degrees of freedom associated with the F-distribution.*

Properties of the F-Distribution

1. The probability curve of the F-distribution is roughly sketched in Fig. 8.3.

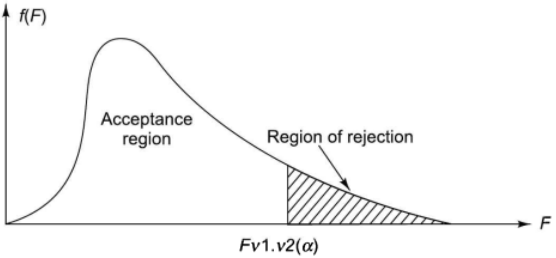

Fig. 8.3

2. The square of the t-variate with n degrees of freedom follows a F-distribution with 1 and n degrees of freedom.
3. The mean of the F-distribution is $\dfrac{\nu_2}{\nu_2 - 2}$ $(\nu_2 > 2)$.
4. The variance of the F-distribution is

$$\frac{2\nu_2^2(\nu_1 + \nu_2 - 2)}{\quad} \quad (\nu_2 > 4)$$

F-test

F-test of significance of the difference between population variances and F-table.

To test the significance of the difference between population variances, we shall first find their estimates, $\hat{\sigma}_1^2$ and $\hat{\sigma}_2^2$ based on the sample variances s_1^2 and s_2^2 and then test their equality. It is known that $\hat{\sigma}_1^2 = \dfrac{n_1 s_1^2}{n_1 - 1}$ with the number of degrees of freedom $\nu_1 = n_1 - 1$ and $\hat{\sigma}_2^2 = \dfrac{n_2 s_2^2}{n_2 - 1}$ with the number of degrees of freedom $\nu_2 = n_2 - 1$.

It is also known that $F = \dfrac{\hat{\sigma}_1^2}{\hat{\sigma}_2^2}$ follows a F-distribution with ν_1 and ν_2 degrees of freedom. If $\hat{\sigma}_1^2 = \hat{\sigma}_2^2$, then $F = 1$. Hence our aim is to find how far any observed value of F can differ from unity as a result of fluctuations of sampling.

Snedecor has prepared tables that give, for different values of ν_1 and ν_2, the 5% and 1% critical values of F. An extract from these tables is given at the end of this chapter. If F denotes the observed (calculated) value and $F_{\nu_1,\nu_2}(\alpha)$ denotes the critical (tabulated) value of F at LOS α, then $P\,\{F > F_{\nu_1,\nu_2}(\alpha)\} = \alpha$.

A sample of size 13 gave an estimated population variance of 3.0 while another sample of size 15 gave an estimate of 2.5. Could both samples be from populations with the same variance? LOS 5%. [$F_{cal} = 1.2$, H_0 is accepted.]

Solution Here, $n_1 = 13$, $\hat{\sigma}_1^2 = 3.0$ and $\nu_1 = 12$, $n_2 = 15$, $\hat{\sigma}_2^2 = 2.5$ and $\nu_2 = 14$.

H_0: $\hat{\sigma}_1^2 = \hat{\sigma}_2^2$, i.e. the two samples have been drawn from populations with the same variance. $H_1 : \hat{\sigma}_1^2 \neq \hat{\sigma}_2^2$.

Let LOS. be 5%.

$$F = \frac{\hat{\sigma}_1^2}{\hat{\sigma}_2^2} = \frac{3 \cdot 0}{2 \cdot 5} = 1.2$$

$$\nu_1 = 12 \quad \text{and} \quad \nu_2 = 14.$$

$F_{0.05\%}$ $(\nu_1 = 12, \nu_2 = 14) = 2.53$, from the F-table. Since $F < F_{0.05}$, H_0 is accepted. That is the two samples could have come from two normal populations with the same variance.

A research was conducted to understand whether women have a greater variation in attitude on political issues than men. Two independent samples of 31 men and 41 women were used for the study. The sample variances so calculated were 120 for women and 80 for men. Test whether the difference in attitude towards political issue is significant at 5% level of LOS? [$F_{cal} = 1.5$, H_0 is accepted.]

In one sample of 9 items, the sum of the squares of deviations of the sample values from the sample mean was 160, and in another sample of 8 observations it was 91. Test whether difference in variance is significant at 5 % level. [$F_{cal} = 1.54$, H_0 is accepted.]

Solution $n_1 = 9, \Sigma(x_i - \bar{x})^2 = 160,$ i.e. $n_1 s_1^2 = 160$

$n_2 = 8, \Sigma(y_i - \bar{y})^2 = 91,$ i.e. $n_2 s_2^2 = 91$

$$\hat{\sigma}_1^2 = \frac{n_1 s_1^2}{n_1 - 1} = \frac{1}{8} \times 160 = 20 ; \quad \hat{\sigma}_2^2 = \frac{n_2 s_2^2}{n_2 - 1} = \frac{1}{7} \times 91 = 13$$

Since $\hat{\sigma}_1^2 > \hat{\sigma}_2^2$, $\nu_1 = n_1 - 1 = 8$ and $\nu_2 = n_2 - 1 = 7$

H_0: $\hat{\sigma}_1^2 = \hat{\sigma}_2^2$ and H_1: $\hat{\sigma}_1^2 \neq \hat{\sigma}_2^2$.

Let the LOS be 5%.

$$F = \frac{\hat{\sigma}_1^2}{\hat{\sigma}_2^2} = \frac{20}{13} = 1.54$$

$F_{0.05}(\nu_1 = 8, \nu_2 = 7) = 3.73$, from the F-table. Since $F < F_{0.05\%}$, H_0 is accepted.

That is, the two samples could have come from two normal populations with the same variance.

We cannot say that the samples have come from the same population, as we

Two independent samples of 8 and 7 items respectively had the following values of the variable:

Sample I	9	11	13	11	15	9	12	14
Sample II	10	12	10	14	9	8	10	

Do the estimates of population variance differ significantly at 5 % LOS?

[$F_{cal} = 1.21$, H_0 is accepted.]

Solution For the first sample, $\Sigma x_1 = 94$ and $\Sigma x_1^2 = 1138$.

$$\therefore \qquad s_1^2 = \frac{1}{n_1}\sum x_1^2 - \left(\frac{1}{n_1}\sum x_1\right)^2$$

$$= \frac{1}{8} \times 1138 - \left(\frac{1}{8} \times 94\right)^2 = 4.19$$

For the second sample, $\Sigma x_2 = 73$ and $\Sigma x_2^2 = 785$.

$$\therefore \qquad s_2^2 = \frac{1}{n_2} \sum x_2^2 - \left(\frac{1}{n_2} \sum x_2\right)^2$$

$$= \frac{1}{7} \times 785 - \left(\frac{1}{7} \times 73\right)^2 = 3.39$$

$$\hat{\sigma}_1^2 = \frac{n_1}{n_1 - 1} s_1^2 = 4.79 \quad \text{and} \quad \hat{\sigma}_2^2 = \frac{n_2}{n_2 - 1} s_2^2 = 3.96$$

since $\qquad \hat{\sigma}_1^2 > \hat{\sigma}_2^2, \nu_1 = 7 \quad \text{and} \quad \nu_2 = 6$

$$H_0: \hat{\sigma}_1^2 = \hat{\sigma}_2^2 \quad \text{and} \quad H_1: \hat{\sigma}_1^2 \neq \hat{\sigma}_2^2$$

$$F = \frac{\hat{\sigma}_1^2}{\hat{\sigma}_2^2} = \frac{4 \cdot 79}{3 \cdot 96} = 1.21$$

$F_{0.05}\,(\nu_1 = 7, \nu_2 = 6) = 4.21$, from the F-table. Since $F < F_{0.05}$, H_0 is accepted.
That is, $\hat{\sigma}_1^2$ and $\hat{\sigma}_2^2$ do not differ significantly at 5% LOS.

t+F test

The nicotine contents in two random samples of tobacco are given below.

Sample 1 : 21 24 25 26 27

Sample 2 : 22 27 28 30 31 36

Can you say that the two samples came from the same population?

Solution

Solution $\bar{x}_1$ = Mean of sample 1 = $\frac{123}{5} = 24.6$

$$\bar{x}_2 = \text{Mean of sample } 2 = \frac{174}{6} = 29.0$$

$$s_1^2 = \text{Variance of sample } 1 = \frac{1}{5}\,\Sigma(x_i - 24.6)^2 = 4.24$$

$$s_2^2 = \text{Variance of sample } 2 = \frac{1}{6}\,\Sigma(x_i - 29.0)^2 = 18.0$$

$$\hat{\sigma}_1^2 = \frac{5}{4} \times 4.24 = 5.30 \text{ and } \nu = 4;\ \hat{\sigma}_2^2 = \frac{6}{5} \times 18.0 = 21.60 \text{ and } \nu = 5$$

$$H_0\colon \hat{\sigma}_1^2 = \hat{\sigma}_2^2; \quad H_1\colon \hat{\sigma}_1^2 \neq \hat{\sigma}_2^2$$

$$F = \frac{\hat{\sigma}_2^2}{\hat{\sigma}_1^2} = \frac{21{\cdot}60}{5{\cdot}30} = 4.07$$

$$F_{0.05}\,(5, 4) = 6.26.$$

Since $F < F_{0.05}$, H_0 is accepted. Therefore, the variances of the two populations can be regarded as equal.

Solution

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\left(\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}\right)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{-4 \cdot 4}{\sqrt{\left(\frac{21 \cdot 2 + 108 \cdot 0}{9}\right)\left(\frac{1}{5} + \frac{1}{6}\right)}}$$

$$= \frac{-4 \cdot 4}{2 \cdot 2943} = -1.92$$

and $\quad \nu = 9.$

From t-table, $F_{0.05}$ $(\nu = 9) = 2.26.$

If H_0: $\bar{x}_1 = \bar{x}_2$ and H_1: $\bar{x}_1 \neq \bar{x}_2$, H_0 is accepted, since $|t| < F_{0.05}$.

That is, the means of two samples (and hence the populations) do not differ significantly. Therefore, the two samples could have been drawn from the same normal population.

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Two random samples gave the following data:

	Size	Mean	Variance
Sample I	8	9.6	1.2
Sample II	11	16.5	2.5

Can we conclude that the two samples have been drawn from the same normal population?

χ^2 test

An unbiased die is thrown 600 times and outcomes are given below. Can we really claim the die is unbiased?

1	2	3	4	5	6
110	98	92	103	105	92

	O_i	E_i	$\frac{(O_i - E_i)^2}{E_i}$

If one of the parent's blood group is A and B then blood group of child is A, AB, B past data shows $A : AB : B = 1 : 2 : 1$. A sample of 250 new born baby with either parent's A is 70, B is 55 and remaining AB. Can we say at 5% LOS population is behaving expectedly?

A	70
AB	125
B	55

H_0 : Population is behaving expectedly
H_1 : Population is not behaving expectedly

	O_i	E_i	$\frac{(O_i-E_i)^2}{E_i}$
A	70	62.5	0.9
AB	125	125	.00
B	55	62.5	0.9
			1.8

$\chi^2_{\text{tab}} = \chi^2_{2,0.05} = 5.9915$

Test statistic:

$\chi^2_{cal} = \sum \left[\frac{(O_i - E_i)^2}{E_i}\right] = 1.8$

Conclusion:

$\chi^2_{cal} < \chi^2_{\text{tab}}$

H_0 is accepted

Population is behaving expectedly.

Question

The following data is collected on two characters. Based on this, Can you say that there is no relation between smoking and literacy?

	Smokers	*Non smokers*
Literates	*83*	*57*
Illiterates	*45*	*68*

H_0 : Literacy & smoking are independent.

H_1 : Literacy & smoking are not independent.

	O_i	E_i	E_i(rounded)	$\frac{(O_i-E_i)^2}{E_i}$
1	83	$\frac{128\times140}{253}=70.83$	71	$\frac{12^2}{71}=2.03$
2	57	$\frac{125\times140}{253}=69.17$	69	$\frac{12^2}{69}=2.09$
3	45	$\frac{128\times113}{253}=57.17$	57	$\frac{12^2}{57}=2.53$
4	68	$\frac{125\times113}{253}=55.83$	56	$\frac{12^2}{56}=2.57$
				$\chi^2_{cal}=9.22$

$\nu=(m-1)(n-1)=(2-1)(2-1)=1$
$\chi^2_{\text{tab}}=\chi^2_{\nu=1,0.05}=3.84$

Test statistic:

$\chi^2_{cal}=\sum\left[\frac{(O_i-E_i)^2}{E_i}\right]=9.22$

Conclusion:

$\chi^2_{cal}>\chi^2_{\text{tab}}$
H_0 is rejected
There is some association between and smoking

Question

Two sample polls of voters for votes for two candidates A and B for a public office are taken, one from among the rural areas and another from urban areas. The results are in the following table

Area	Votes for	
	A	B
Rural	620	380
Urban	550	450

Examine whether the nature of the area is related to voting preference in this election.